**(iii) Verification of Fourier Transform Properties and Frequency-Domain Characteristics.**

**Software / Laboratory Environment:**

MATLAB

**MATLAB Coding:**

```
clc;
clear;
close all;

% Experiment 7(iii): Verification of Fourier Transform Properties

t = -3:0.001:3;
dt = t(2)-t(1);
t0 = 0.4;

x = exp(-t.^2);

% Time-shifted signal
x_shift = interp1(t,x,t-t0,'linear',0);

% FFT with zero-padding
N = 8192;
X = fftshift(fft(x,N))*dt;
X_shift = fftshift(fft(x_shift,N))*dt;

% Frequency axis
f = (-N/2:N/2-1)/(N*dt);

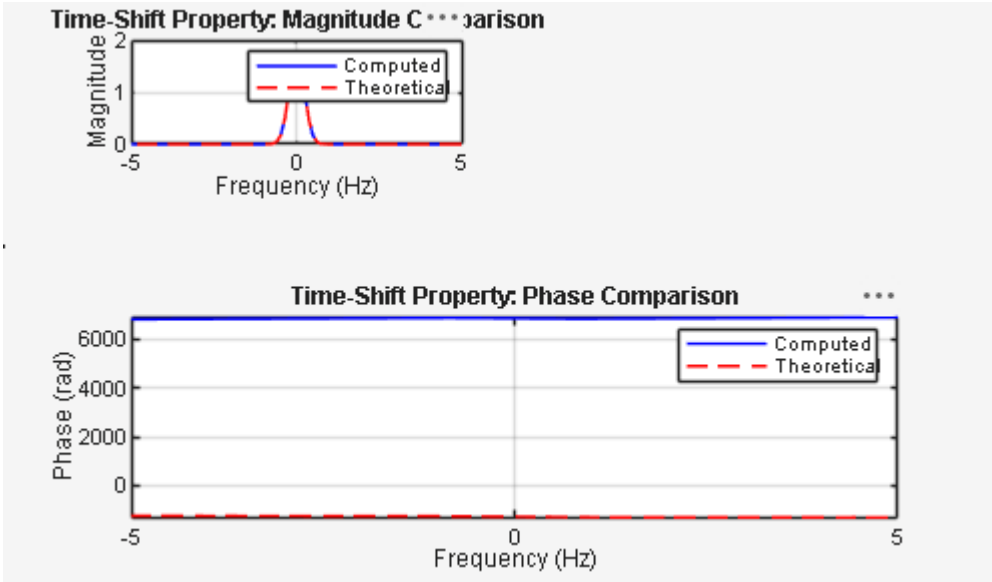
% Theoretical time-shift property
X_shift_theory = X.*exp(-1j*2*pi*f*t0);

% Magnitude comparison
figure;
```

```
subplot(3,2,1);
plot(f,abs(X_shift),'b',f,abs(X_shift_theory),'r--','LineWidth',1.1);
xlim([-5 5]);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Time-Shift Property: Magnitude Comparison');
legend('Computed','Theoretical');

% Phase comparison
subplot(2,1,2);
plot(f,unwrap(angle(X_shift)),'b', ...
    f,unwrap(angle(X_shift_theory)),'r--','LineWidth',1.1);
xlim([-5 5]);
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (rad)');
title('Time-Shift Property: Phase Comparison');
legend('Computed','Theoretical');
```

**Output Graphs:**



**RESULT:**

The continuous-time Fourier transform of representative signals was successfully obtained using FFT-based approximation, and their magnitude and phase spectra were analyzed to identify significant frequency components. The time-shift property was verified by comparing the computed and theoretical Fourier transforms, confirming that time shifting preserves magnitude while introducing a frequency-dependent phase shift. The spectral components and occupied bandwidth of a composite communication signal were also determined, demonstrating the importance of frequency-domain analysis in communication-system design.

**RUBRICS:**

| **Objective** | **Assessment Criterion** | **Excellent – 4** | **Good – 3** | **Satisfactory – 2** | **Needs Improvement – 1** | **Weightage** | **Marks(10)** | **Marks obtained** |
|---|---|---|---|---|---|---|---|---|
| **(ii) To determine the magnitude and phase spectra of continuous-time signals and analyze their significant spectral** | **Fourier Transform Computation & MATLAB Implementation** | Computes X(jω) of every given CT signal correctly, analytically and numerically . Uses a suitable time range, sampli | Computes the spectra correctly, with minor errors in scaling, sampling rate, frequency axis or parameters. | Computes the spectra for only some signals, or has noticeable errors in code, scaling or frequency axis, and needs assista | Unable to compute the Fourier transform or produce meaningful spectra. | **20** | 2 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **components.** | | ng rate and frequency axis in rad/s or Hz. | | nce. | | | | |
| | **Magnitude & Phase Spectrum Plots** | Plots $\|X(j\omega)\|$ and $\angle X(j\omega)$ correctly for every signal, with proper scaling, titles, axis labels and grid. Handles phase wrapping and phase near zero-magnitude points correctly. | Plots both spectra correctly, with minor errors in scaling, labelling, phase wrapping or axis range. | Plots the spectra with noticeable errors in scaling, phase handling or labelling, and needs guidance. | Unable to plot the magnitude and phase spectra. | **20** | 2 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | **Spectral Component Analysis** | Correctly identifies dominant frequencies, peaks, bandwidth and the frequency range holding most of the signal energy. Uses symmetry (even magnitude, odd phase for real signals) and relates the spectral lines or lobes | Identifies most significant spectral components and bandwidth, with minor errors or limited justification. | Identifies only basic spectral components. Bandwidth or symmetry reasoning is incomplete, and needs guidance. | Unable to identify or justify the significant spectral components. | **20** | 2 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | to the signal parameters. | | | | | | |
| | **Analytical vs Numerical Verification** | Compares the analytical and MATLAB spectra, quantifies the difference and explains its causes such as truncation, sampling and spectral leakage. Shows how signal parameters change | Compares analytical and numerical results correctly, with brief explanation of differences. | Makes a basic comparison with noticeable errors, and the explanation of differences is incomplete. | Unable to verify the numerical spectra against the analytical results. | **20** | 2 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | the spectrum. | | | | | | |
| | **Engineering Analysis, Record & Interpretation** | Provides technically accurate interpretation of the results and relates them to the physical or engineering meaning of the signal. Maintains a complete, well-organized record with | Maintains a proper record with correct interpretation, with minor omissions in explanation or conclusions. | Record and interpretation are partial or only descriptive, and need guidance. | Incomplete record, limited or incorrect interpretation of the results. | **20** | 2 | |

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|---|---|---|---|---|---|---|---|---|
| | | observations, tables, plots and conclusions. | | | | | | |
| | | | | | **Total** | **100** | **10** | |